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Original article

Accuracy assessment of noise prediction of propellers via numerical simulation, analytical methods, and experimental campaigns

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ABSTRACT

This work focuses on the assessment of the accuracy of numerical predictions and experimental campaigns on providing aerodynamic characteristics and acoustic emissions of an isolated benchmark propeller. An experimental campaign is carried out with a benchmark model low-Reynolds propeller (89,000 tip Reynolds number) of 0.3 m diameter operating at high RPM, equivalent of a tip-Mach number (M_t) of 0.37 and an advance ratio (J) of 0.4. Measurements are conducted in the anechoic A-Tunnel, an open test-section wind tunnel, confined by an anechoic chamber. Simulations are carried out with the commercial software PowerFLOW and aimed at reproducing the propeller geometry and conditions. BEMT-based noise estimations are also carried out and used as reference results for the discussion. The study focuses on the assessment of uncertainties of the experimental campaign, and the current accuracy of numerical and analytical predictions, creating a complete picture of the discrepancies expected in similar setups when predicting and experimenting with propeller noise along with the potential sources of errors. Results point to the accuracy of the three methodologies in assessing the overall noise emissions. This follows the precise description of the integral aerodynamic loads on the propeller and the dominant first tonal components of the noise signature. Similarly, the velocity field obtained from the methodologies compares well. Nevertheless, precise descriptions and measurements of the higher harmonics of the tonal emissions and of the broadband noise levels are still lacking and demonstrate the need for improved experimental and numerical methodologies to allow accurate cross-validation between the methodologies.

1. Introduction

The application of aerial vehicles in urban environments is seen as a potentially viable commercial solution for last-mile transport of parcels and even transport of passengers [1]. Vehicles designed for both applications have in common electric motors and rotors, most of them consisting of a combination of open propellers. The latter often brings advantages of propulsive efficiency, necessary for the usage of batteries as energy storage. However, propellers are also responsible for significant noise emissions, which are considered one of the key factors that will enable or block the application of aerial vehicles in urban environments [2].

Therefore, the prediction of noise and design of efficient and silent propellers has been the focus of attention of many designs and academic studies [3–5]. The spectrum of noise emitted by propellers is, in general, composed of two components, i.e. a tonal and a broadband one [6]. Tonal noise is observed on the harmonics of the blade-passing frequencies (BPF) and is the loudest at the first of the harmonics. Broadband noise is generated by the airfoil self-noise, turbulent boundary layer trailing edge noise [7,8], scaling with the boundary layer thickness, or by turbulence impingement noise, which is dependent on the intensity and scales of the turbulent fluctuations in the atmosphere [9,10]. In most typical conditions of applications, high tip speeds, and Reynolds numbers, the tonal component of noise dominates the overall emissions from an isolated propeller. Specifically, the first harmonics of the BPF encompass almost the entirety of the acoustic energy for steady loading conditions. However, for small drones and urban air mobility vehicles, this figure can be significantly different. The low-Reynolds number of operations of the propellers of small drones and the potential turbulent flow field in which they fly cause the relevance of broadband components to rise. Besides, the low rotational speeds of the large propellers for urban air mobility concepts and installation effects decrease the

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influence of the first tonal components while increasing the importance of the higher harmonics of the blade passing.

In this scenario, the relevance of the highest harmonics of the tonal noise and broadband noise increases. Thus, accurate prediction, modelling, and experimentation of tonal, in all its harmonics, and broadband noise from propellers are fundamental for the correct interpretation of noise emissions from novel urban air transportation systems. In that regard, several works have been dedicated to the assessment of noise from isolated propellers. Casalino et al. (2021) [11] first proposed a benchmark propeller geometry and accompanying experiments describing the flow and acoustic properties of the propeller. Their results have been later used for the assessment of numerical simulations, yielding good agreement with broadband and tonal components [12]. Casalino et al. (2023) [5] have shown that a better match of flow field and noise emissions is obtained when the recirculation in the room is also considered. Nevertheless, discrepancies still exist and are mostly attributed to experimental uncertainties, e.g., manufacturing capability [13], motor and background noise, and numerical model approach [14]. The work of Domogalla et al. (2026) [15] has demonstrated agreement between low-fidelity methodologies and flyover measurements of a Do-228. Still, discrepancies exist and, for the overall sound pressure level, are in the order of 5 dB. Similarly, the work of Bergmann et al. (2022) [16] has compared BEMT-based with numerical-based simulations and free-field experiments. Results have pointed to large discrepancies between methods, both in levels and in directivity patterns.

This research is dedicated to a discussion on the currently achievable accuracy of experimental, numerical, and analytical techniques for evaluating the noise of an isolated propeller. A dedicated experimental campaign was carried out with a benchmark propeller, followed by equivalent blind analytical prediction and numerical simulations. These analyses are placed together in this work to create a comparison between the three methodologies, carried out independently. The levels of tonal and broadband noise accuracy captured from the methodologies are used to assess the most important sources of noise and the best practices from isolated propellers and the sources of possible discrepancies between the different approaches. A description of the model and flow conditions is shown in Section 2 followed by a description of the experimental, numerical, and analytical procedures (Sections 3–5 respectively). The results and discussions (Section 6) are devoted to the comparisons between the three methodologies for the aerodynamic and acoustic characteristics predicted. The work concludes with a final remark on the outcomes from the exercise together with future guidelines and observed limitations from each of the methodologies assessed.

Table 1
Conditions of the tested propeller.

Parameter	Symbol	Value
Propeller diameter (m)	D	0.30
Propeller number of blades	N	2
Propeller rotational speed (RPM)	n	8,000
Flow speed (m/s)	U_∞	16.0
Advance ratio	$J = \frac{60U_\infty}{D.n}$	0.40
Tip Mach number	M_t	0.37
Tip Reynolds number	Re_t	89,000
Blade Passing Frequency (Hz)	BPF	267

2. Model and flow condition

The work is a blind test study for the assessment of modelling and experimentation of an isolated propeller. This means that numerical, experimental, and analytical techniques are carried out independently by each of the authors without any prior information about the results obtained by each other. For this purpose, a benchmark propeller model is selected for the tests. The model propeller is based on an APC 9×6 propeller scaled to 0.3 m diameter, using a NACA 4412 airfoil geometry. The propeller geometry is described in the work of Grande et al. (2022) [17] (see Fig. 1).

Measurements are carried out with the propeller at 8000 rotations per minute (RPM) and with 16 m/s flow speed, corresponding to an advance ratio of 0.4 and tip Mach number of 0.37. Table 1 summarizes the flow conditions adopted for the campaigns. The choice of conditions is deemed more representative of real applications when compared to previous validations carried out at smaller tip Mach numbers [4]. Besides, these conditions allow for improved acoustic measurements at the anechoic wind-tunnel facility, following the cut-off frequency of its anechoic treatment [18]. The high-pitch propeller geometry selected also yields non-ideal flow conditions at low advance ratios, with more controlled conditions reflected at the selected advance ratio, i.e. reduced presence of separation bubble and consequent influence of transition from laminar to turbulent flow on the comparisons. For the experiments, the transition is not forced by means of tripping devices, since the ability to predict the natural transition is also deemed relevant for the assessment of numerical methodologies. Alongside the propeller geometry, these are the only pieces of information shared a priori to the tests and simulations conducted. The focus of this work is to discuss the achievable accuracy of numerical, modelling, and experimental endeavours fully dedicated to a single condition. The choice of geometry and flow

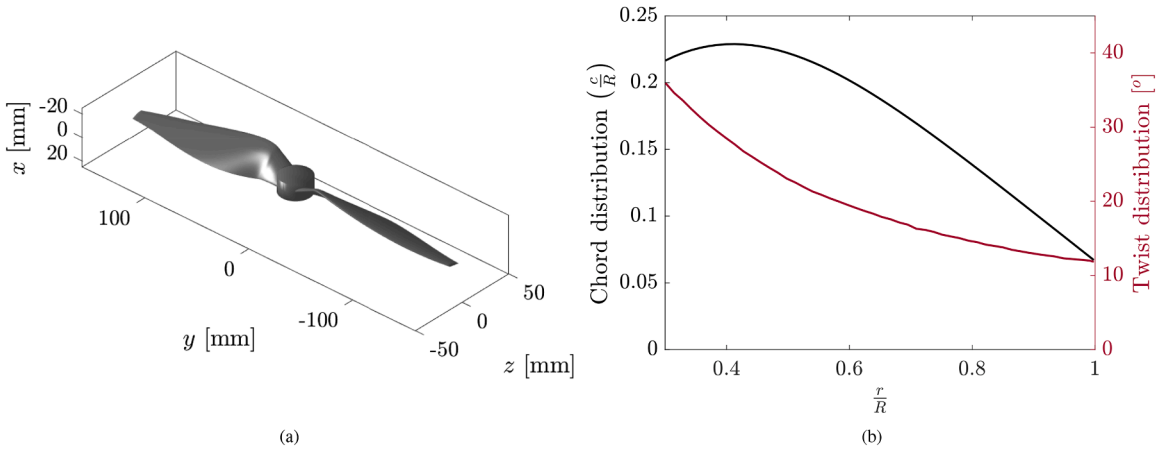
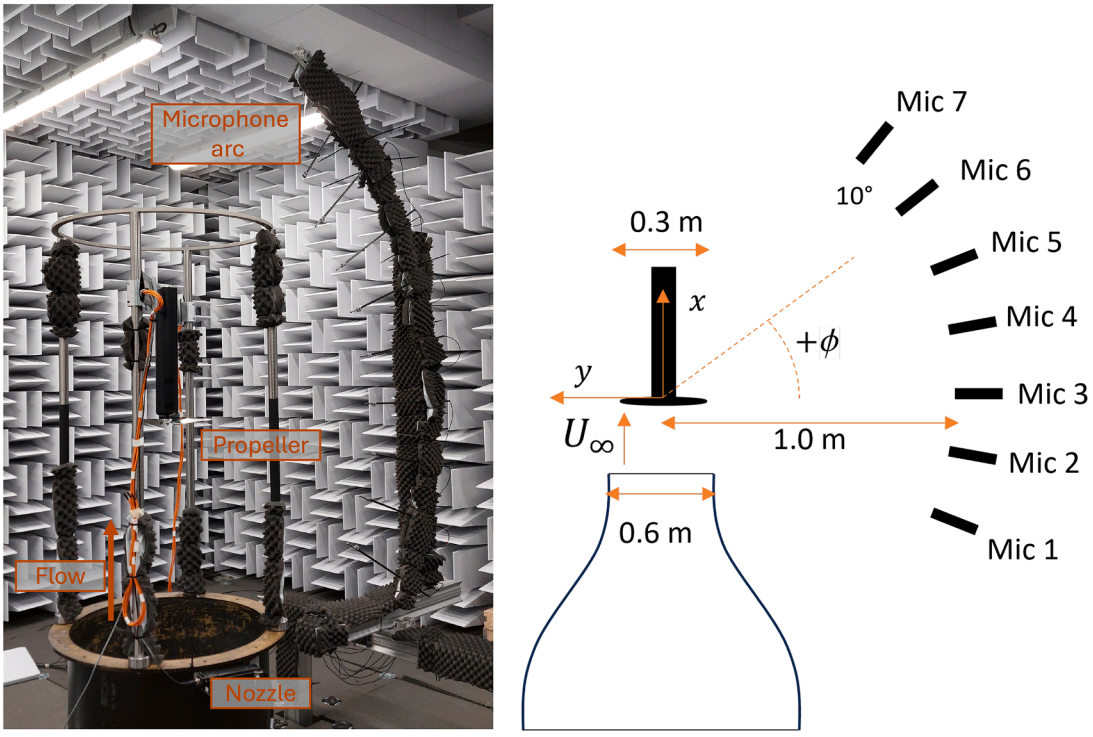


Fig. 1. Propeller geometry, chord and twist distribution along the radius (r).



(a) (b)

Fig. 2. Experimental setup (a) and geometric features of relevance for the study conducted (b).

conditions can significantly affect the quality of the comparisons. Exploring this, however, is deemed out of the scope of this work.

3. Experimental campaign

3.1. Wind-tunnel model and flow facility

The propeller is driven by a Leopard Hobby 3536-5T 1520 KV motor. The motor is attached to a 6-axes load cell by Futek, model mini 40 with limits of 120 N of thrust and 2 Nm of torque. A Digital EM1 differential optical encoder is used to control the propeller RPM and reference the propeller position during the acoustic acquisition. The whole setup is encapsulated into a faired propeller rig, which minimizes flow interference and spurious noise sources. The propeller is mounted 0.3 m from the circular nozzle exit with 0.6 m diameter. At this location, the estimated shear-layer development, following the works of [19,20], should not affect the core flow in more than 1% within a diameter of 0.5 m (1.67D) from the jet centre.

Experiments are conducted in the A-Tunnel facility of Delft University of Technology [18]. The open test section wind tunnel encompasses an anechoic room of $6.4 \times 6.4 \times 3.4$ (height) m^3 designed to absorb acoustic waves above 150 Hz. Flow speeds of up to 34 m/s can be achieved with turbulence intensities below 0.1%. Fig. 2 shows the propeller model mounted in the wind-tunnel facility.

3.2. Particle image velocimetry (PIV) measurements

Besides the integral loads, PIV measurements are also conducted on a plane parallel to the flow direction. The setup follows a standard stereo-PIV application [21] using low-speed LaVision Imager sCMOS cameras (16 bit image pair, 2560×2160 pixels, 10 μs pixel-pitch, minimum 120 ns interframe time) as sensors, a Quantel EverGreen 200 laser (200 mJ per pulse, 15 pulses per second) as light source, and SAFEX oil particles for seeding (0.5 μm median particle, 1 μs response time). A total of 1000 images are captured with a 20 μs pulse separation, equivalent to 1.0 degrees of blade rotation. The images are post-processed, calibrated, and cross-correlated with DaVis 10 software from LaVision GmbH. Triggering signals are acquired simultaneously with the acoustic and the encoder information, allowing for later retrieval of the blade phase angle. Table 2 summarizes the main characteristics of the PIV technique employed.

Table 2
Properties of the PIV technique employed during the experimental campaign.

Parameter	Value
Field of view ($X \times Y$)	0.36×0.18 m ($1.2 \times 0.6D$)
Vector resolution	0.34 mm $0.1\%D$
Velocity uncertainty	0.8 m/s $5\%U_\infty$
Number of frames	1000

3.3. Acoustic measurements

Acoustic measurements are carried out with a microphone arc consisting of 7 microphones, placed at 1 m from the propeller centre of rotation and describing azimuthal angles (ϕ) from −20 to 40 ° with 10° step. Fig. 2 shows the microphone arc, the propeller setup, and the geometric description of the experiment.

The acoustic acquisition is controlled by a PXI Express system with microphones connected to PXIe-4499 boards with 24-bit precision. Acoustic acquisitions are taken for a duration of 30 s with a sampling rate of 102,400 samples per second. The microphones used are GRAS 46BE with flat response from 10 to 40 kHz (max 1 dB variation quoted from manufacturer). The calibration of the microphones is carried out in situ using a GRAS 42AG pistonphone operating at 1,000 Hz and 94 dB of sound pressure. Further details on the accuracy of the acoustic setup used are shown in Appendix A.

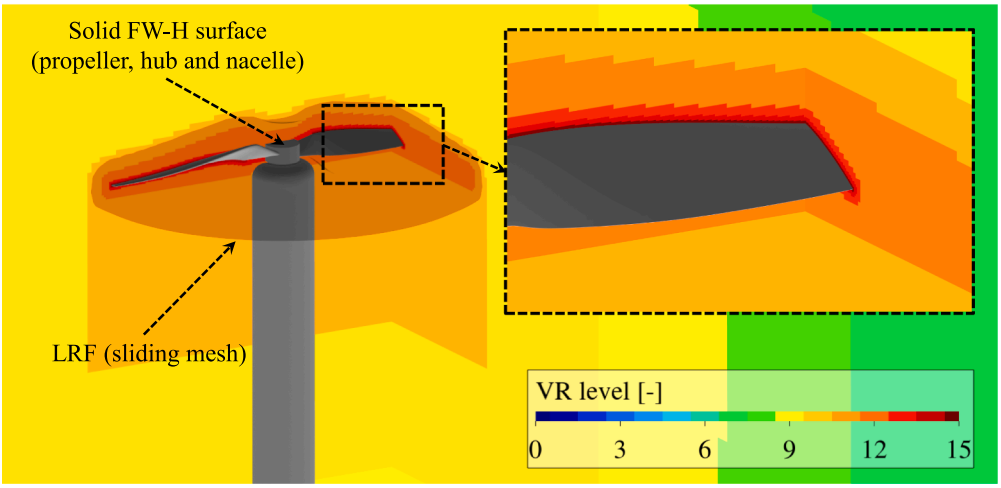


Fig. 3. Sketch of computational setup and near body mesh.

4. Numerical simulation

The numerical simulation is carried out using the Lattice-Boltzmann/Very-Large Eddy Simulation (LB/VLES) solver SIMULIA PowerFLOW. The numerical setup is based on that employed in previous studies by Casalino *et al.* and uses a recently developed variant of the VLES model capable of predicting laminar to turbulence natural transition phenomena [4,5]. The computational fluid domain is a spherical volume of 325D radius, centred around the propeller hub. Freestream static pressure and velocity, and turbulence intensity of 0.1% of the freestream velocity, are prescribed on its outer boundary. An anechoic environment is reproduced in the numerical computations through an acoustic sponge that is used to dissipate the out-going acoustic waves and minimize the backward reflection from the outer boundary. The sponge is defined by two concentric spheres of radius 15D and 55D, respectively, centred around the propeller hub. The fluid kinematic viscosity is gradually increased starting from its physical value within the inner sphere, up to an artificial value two orders of magnitude higher outside the outer one.

Details of the computational setup and grid in proximity to the propeller geometry are shown in Fig. 3. The propeller and hub are encompassed by a volume of revolution that defines the Local Reference Frame (LRF), which is the sliding mesh used to reproduce the propeller rotation. A total of 16 Variable Resolution (VR) regions are used to discretize the whole fluid domain. A resolution of 400 voxels along the mean chord value of 22.85 mm is used in VR13, resulting in a smallest voxel size of 0.0571 mm and an average y^+ over the propeller blade below 10. Flow confinement effects are not considered following the results of Casalino et al. (2023) [5], which demonstrated weak dependency on the effect for high advance ratios.

The whole fluid domain is initialized with the instantaneous flow solution from a statistically converged coarser simulation. Hence, after a settling time corresponding to 2 propeller revolutions, the sampling of relevant flow data is started for 50 additional revolutions. The aerodynamic noise generated by the propeller is evaluated by means of a time-domain Ffowcs-Williams Hawkings (FWH) approach [22] based on Farassat's formulation 1A [23] applied to the propeller, hub, and nacelle surface pressure. Acoustic data are sampled at 185 kHz with a spatial averaging of about 0.5 mm on the solid FWH integration surface.

5. Blade element momentum theory (BEMT) based noise estimation

For comparisons with expected results from the theory of propeller noise, BEMT-based noise estimation is also carried out. The BEMT-based noise estimations follow the works of [11,24], carried out with the software *opty*-dB *pnoise*. The flow model is based on a conventional implementation of the BEMT formulation for uniform inflow and includes Prandtl tip-losses. Lift and drag coefficients from the airfoil sections are obtained from a potential flow solver coupled with a boundary layer model, following [25]. Transition location is predicted following a simplified e^n-method, analogous to the X-Foil software.

From the aerodynamic force distribution, the tonal sources are estimated based on a time-domain description of moving compact distributed dipole sources for the load noise and monopole ones for the thickness noise [24]. Broadband noise estimations follow the work of Roger & Moreau [26], adapted for a rotating framework [27]. The turbulent boundary layer wall-pressure spectra required for the estimations are also obtained using Schlinker's model [27]. The latter choice follows the simplicity and robustness of the model. Other wall-pressure spectrum models require detailed data of the boundary-layer state near the trailing-edge, which is not always straightforward to retrieve. Besides, most models rely on semi-empirical corrections, and their suitability remains case-dependent. Thus, as this work proposes a blind test with minimal a priori knowledge, Schlinker's model is deemed the simplest option. Future work can explore possible improvements from other available semi-empirical models [28–32].

6. Results and discussions

6.1. Aerodynamic comparisons

Comparisons of the flow field and total aerodynamic forces are shown to support the further detailed analysis of the emitted acoustic field from the propeller. Table 3 shows the obtained thrust, torque, and power from each of the methodologies. Comparisons throughout this work will always be referenced with respect to theoretical, analytical estimations.

Analyses of the thrust level results reveal the small deviations between the three methodologies, with variations of thrust not higher than 0.1 N, representative of less than 5% of the thrust coefficient. The predicted thrust is the smallest for the numerical method, with the BEMT methodology providing the highest of them.

This, however, is the opposite of the analysis of the required torque and power, where the numerical estimations predict the highest required torque and, thereby, power. The experimental campaign results in a significantly lower required torque and power, around 5% below the analytical predictions. This can be associated with the transition prediction for the BEMT model (threshold N factor) and turbulence intensity quoted for the simulations, which depend on facility conditions and vi-

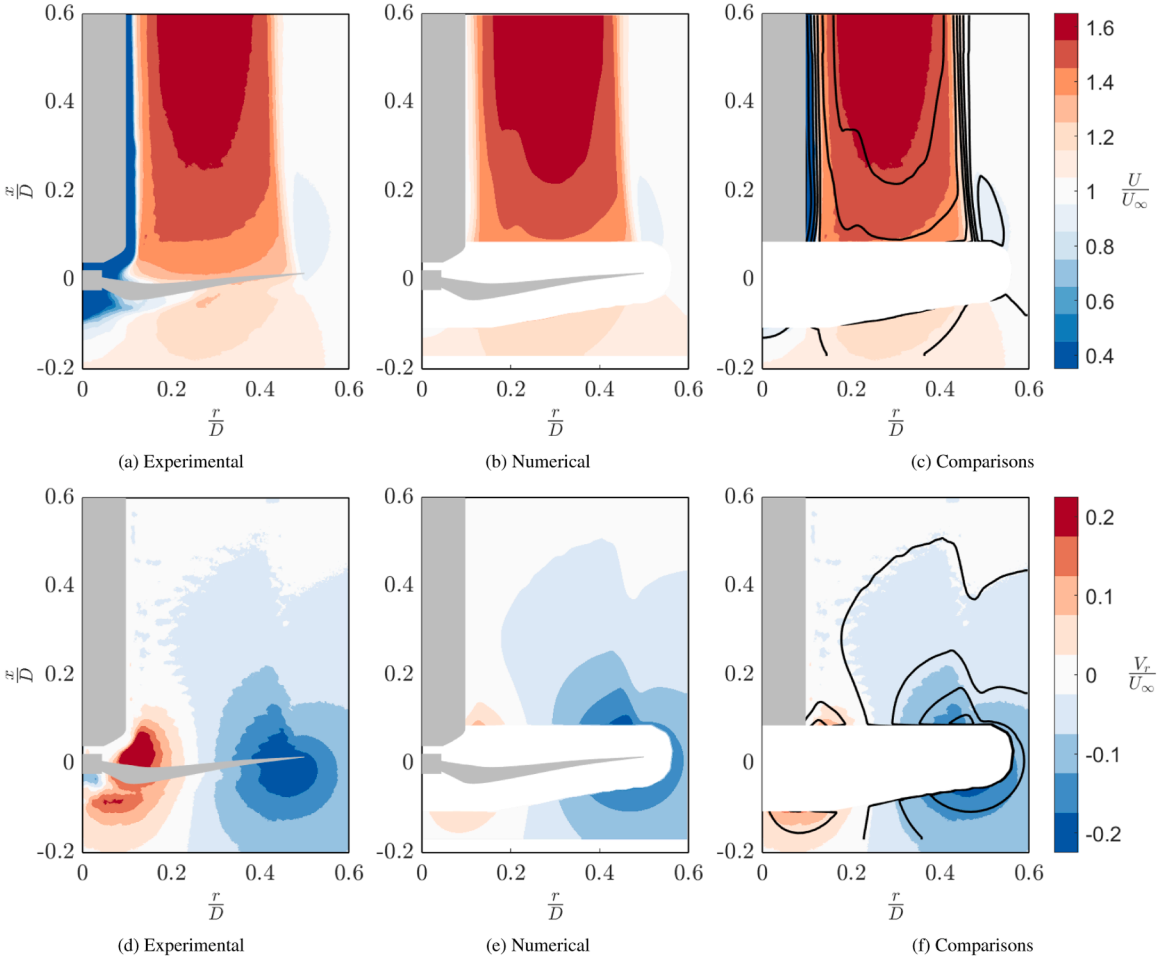


Fig. 4. Comparison of the average flow field at a plane orthogonal to the propeller rotation direction. From left to right the figures show the experimental data, the numerical data, and the comparison among them. On the latter, colours represent the experimental information and the contour lines show the frontier of the same levels obtained from the numerical simulations. The first row shows the axial velocity field and the second the radial velocity field. Whited areas in the numerical data represent the boundaries of the rotating frame.

Table 3
Thrust, torque, power, thrust coefficient, and power coefficients obtained from each of the three methodologies employed.

	BEMT	Numerical		Experimental	
	Value	Value	Deviation	Value	Deviation
Thrust (N)	14.56	14.51	−0.05	14.55 ±0.01	−0.01
Torque (Nm)	0.4338	0.4361	+0.0023	0.4133 ±0.0002	−0.0205
Power (W)	363.4	365.4	+2.0	346.2 ±0.1	−17.2
$C_t = \frac{T}{\rho n^2 D^4}$	0.0826	0.0822	−0.0004	0.0825 ±0.00006	−0.0001
$C_p = \frac{P}{\rho n^3 D^5}$	0.0512	0.0518	+0.0006	0.0491 ±0.00002	−0.0021

bration levels [18]. The good agreement demonstrated for the integral loads on the propeller indicates that the three methodologies are equally suitable for the assessment of the aerodynamic performance of the propeller and conditions are minimally equivalent for the assessment of the acoustic emission from all the methods.

Following the PIV measurements, the flow field on a plane parallel to the axial direction of the propeller can be compared. Here only numerical and experimental data are shown. Fig. 4 shows the average flow field obtained for both experimental (leftmost figure) and numerical (middle figure). The plot on the right shows the overlap between the numerical and the experimental flow fields. In the latter, colour shades show the experimental data while the isocontour lines the equivalent values for numerical data. The top row shows the average axial flow field (U) while the bottom one shows the radial velocity (V_r). The surface of the rotating frame used for the computations is shaded and should be disregarded for the comparisons.

The results demonstrate the agreement expected from the integral loads. Deviations are hardly below 5% of the free-stream velocity. Discrepancies are stronger in the regions close to the edges of the testing rig, caused by reflections from the rig. The overall characteristics of the flow field are well captured by both techniques with slightly stronger stream-tube contraction observed in the experiments in comparison to the numerical simulations.

Similar comparisons are also seen for the phased-averaged flow field obtained with the numerical and experimental data. Fig. 5 shows, in the same fashion as the previous figure, the flow field obtained as the propeller blade passes through the flow field plane. The top row demonstrates the phase-averaged streamwise velocity while the bottom one demonstrates the out-of-plane vorticity field. Again, the image demonstrates the agreement between the two methodologies. The largest errors are in the order of a single shading, which is smaller than 10% of the free stream velocity and $10D/U_\infty$ for the vorticity field. The reflections, and consequent uncertainties, of the PIV field close to the wall of the propeller rig do not allow for the observation of the opposite vortex core formed along the propeller root. The core vortex is shown more inwards and downstream on the experimental measurements with re-

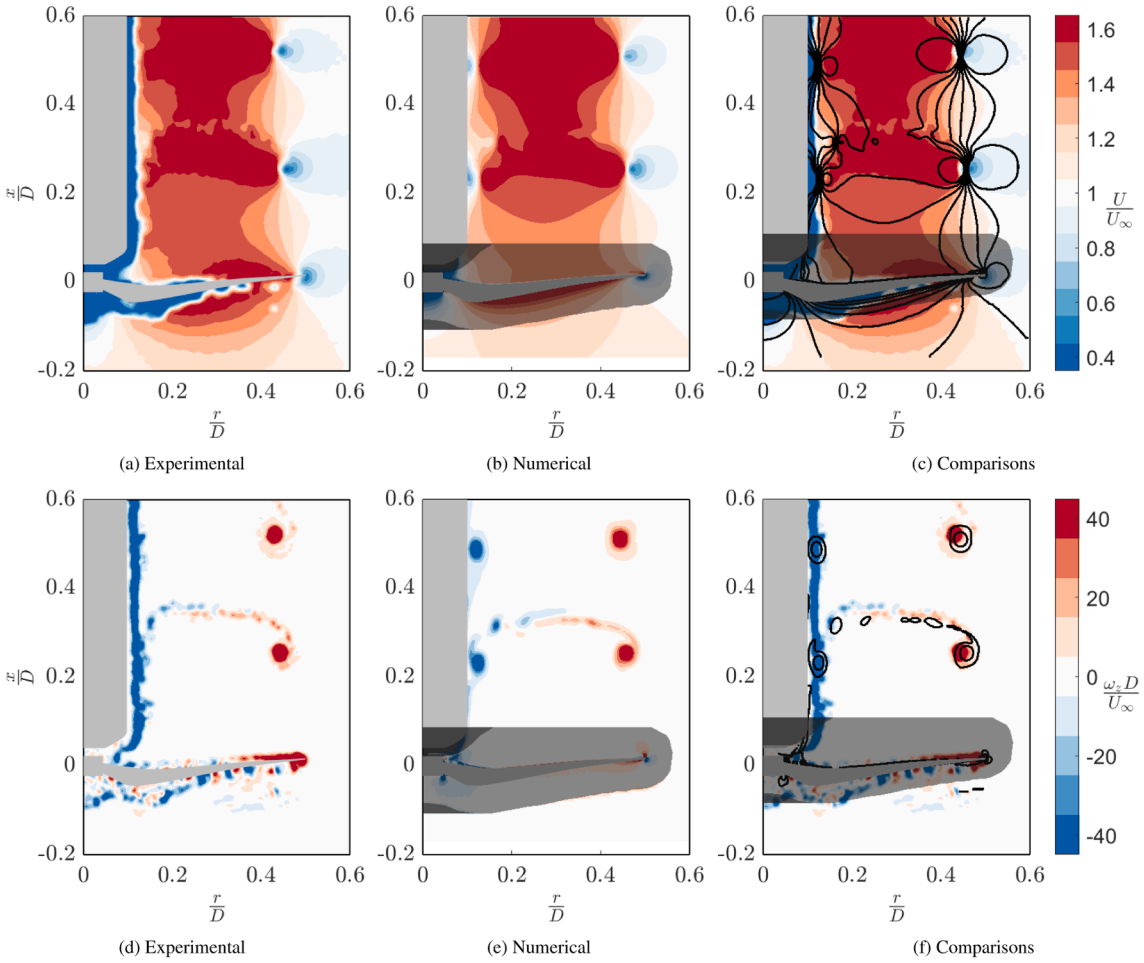


Fig. 5. Comparison of the phase-averaged flow field at a plane orthogonal to the propeller rotation direction. From left to right the figures show the experimental data (taken from 10 images of the propeller passing through the measurement plane), the numerical data (taken from the average of the 50 revolutions), and the comparison among them. On the latter, colours represent the experimental information and the contour lines show the frontier of the same levels obtained from the numerical simulations. The first row shows the axial velocity field and the second the out-of-plane vorticity field. Shaded areas in the plots containing numerical data represent the boundaries of the rotating frame.

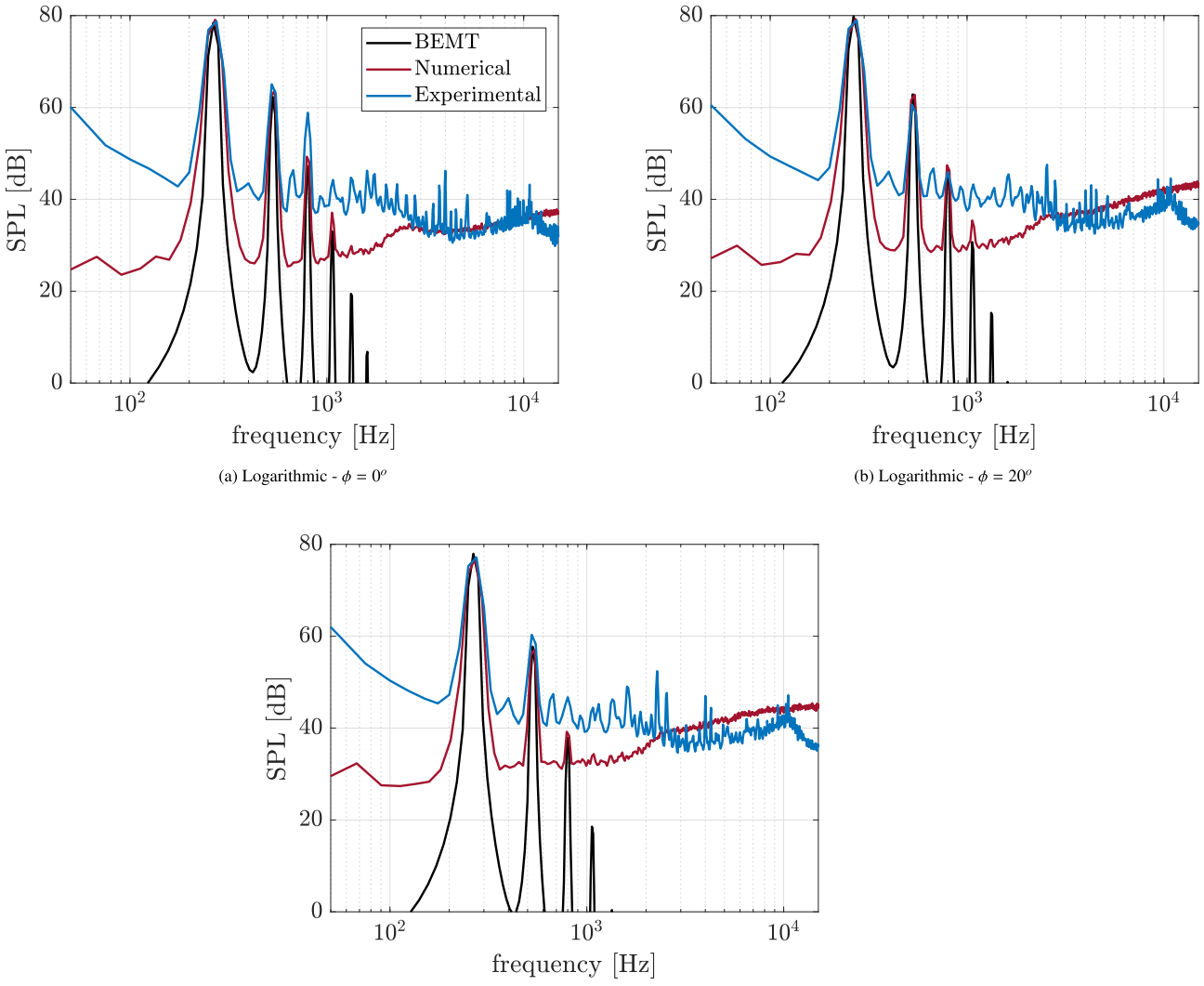
spect to the numerical one. This follows the slightly higher thrust and stream-tube contraction observed in Table 3 and Fig. 4 and could be driven by small differences in the flow speed between the two methodologies. The higher thrust can be associated with the ideal numerical conditions, where effects due to recirculation, nozzle and room geometry were neglected for the numerical simulations. Previous works have demonstrated the possible impact of these considerations on the flow field [5].

In conclusion, the observations and comparisons of the flow field and integral loads demonstrate that all three methodologies are capable of accurately capturing the aerodynamics of the propeller. Deviations in thrust and torque do not exceed 5% of the total values. Flow-field comparisons between numerical and experimental techniques have also revealed small differences for both average and phase-locked flow fields. The results demonstrate the agreement between the three methodologies in characterizing the aerodynamic field over the propeller. These analyses support the similarity of the flow field between all compared methodologies and allow for a meaningful comparison of the acoustic field, measured or predicted, of the propeller. Still, the analyses presented here do not allow for a meaningful assessment of turbulence statistics, spectral distribution, or small flow asymmetries. These quantities are fundamental for the study of broadband noise and high-harmonics of the blade passing frequency, which will be explored in the coming parts of this study.

6.2. Comparisons of tonal noise

Before a detailed analysis of the acoustic emissions and their estimations by different methodologies, Table 4 summarizes the Overall Sound Pressure Level (OSPL), with and without A-weighting, estimated by the three methodologies. Results clearly point to the high accuracy of all methodologies in predicting the overall level. Level discrepancies are no larger than 1.5 dB and higher levels are shown for both the numerical simulations and the experimental data. Besides, the good predictions for both non-weighted and A-weighted noise emissions point to a reasonable estimation of the spectral distribution from all methodologies.

Following the overall sound level estimations, the spectrum of the sound pressure level (SPL) for tonal predictions and experimental results is demonstrated in Fig. 6 at different azimuthal angles ϕ for all techniques. The results are presented here in SPL for direct comparison of the tonal components, i.e., independent of the frequency band of each dataset. The SPL is obtained from the integration of the power spectral density, obtained from Welch's method, along the frequency band, taken here as the same band as the one used to obtain the power spectral density. Nevertheless, the experimental, numerical, and analytical spectra are obtained with a frequency discretization of 25, 22.6, and 15 Hz, respectively, and comparisons of the broadband signal are also fairly possible (0.4 dB lower for the numerical case in comparison



(a) Logarithmic - $\phi = 0^o$

(b) Logarithmic - $\phi = 20^o$

(c) Logarithmic - $\phi = 40^o$

Fig. 6. Comparison of the experimental, numerical, and BEMT-based acoustic spectrum of the propeller at different azimuthal angles (ϕ). BEMT-based predictions only include the tonal components.

Table 4
Overall sound pressure level at $\phi = 0°$ azimuthal angle predicted by each methodology along the plane of the propeller rotation. Data is shown both without and with A-weighting applied.

	BEMT	Numerical		Experimental	
	Value	Value	Deviation	Value	Deviation
OSPL (dB)	83.4	84.7	+1.3	84.6	+1.2
OSPL (dBA)	76.2	76.8	+0.6	77.3	+1.1

with the experimental one). The broadband part of the spectra of the analytical model is omitted in the following comparisons.

As observed, the numerical simulation and analytical model represent fairly well the level of the first BPF harmonic of the propeller noise in comparison to the experiments, with the measured tone being 0.2 dB higher than the predicted one from the BEMT and 0.3 dB from the numerical one. This follows the accuracy of the overall sound pressure estimations by all methodologies. Nevertheless, as the harmonic number is increased, and levels are reduced, the associated error also increases. The second harmonic of the BPF shows larger discrepancies, with the measurements demonstrating a tone 2.7 dB higher than the BEMT predictions and 1.5 dB higher than the numerical ones. Both numerical and analytical techniques demonstrate the expected fast decrease of the tonal levels characteristic of steady loading noise. The higher harmonics demonstrate much higher uncertainties. To these, experimental sources of spurious noise are mostly driving the differences. Electric and mechanical noise from the experimental setup are possibly hampering the comparison of the 3rd, 6rd, and 7rd BPF harmonic number. Appendix A shows a detailed analysis of the experimental uncertainties associated with the acoustic measurements. Amongst BEMT and numerical data, discrepancies are in the order of 0.7 dB for the azimuthal angle $\phi = 0^o$.

The phase-lock averaged time history of pressure along one full rotation of the blade is here shown for comparing BEMT-based predictions, experimental, and numerical results.

Fig. 7 shows the resulting average time history of the acoustic pressure along one full propeller rotation. The pressure fluctuations are obtained by the phase-average of the acoustic pressure signal along the blade rotation. This is made possible during the experiments by synchronously acquiring the acoustic data and the signal of the encoders [33]. By carrying on this process, the tonal component of the noise is separated from the broadband one. Phase averages are made with the 4000 rotations from the experimental data and the 50 rotations. The suppression of the broadband noise from this technique follows the in-

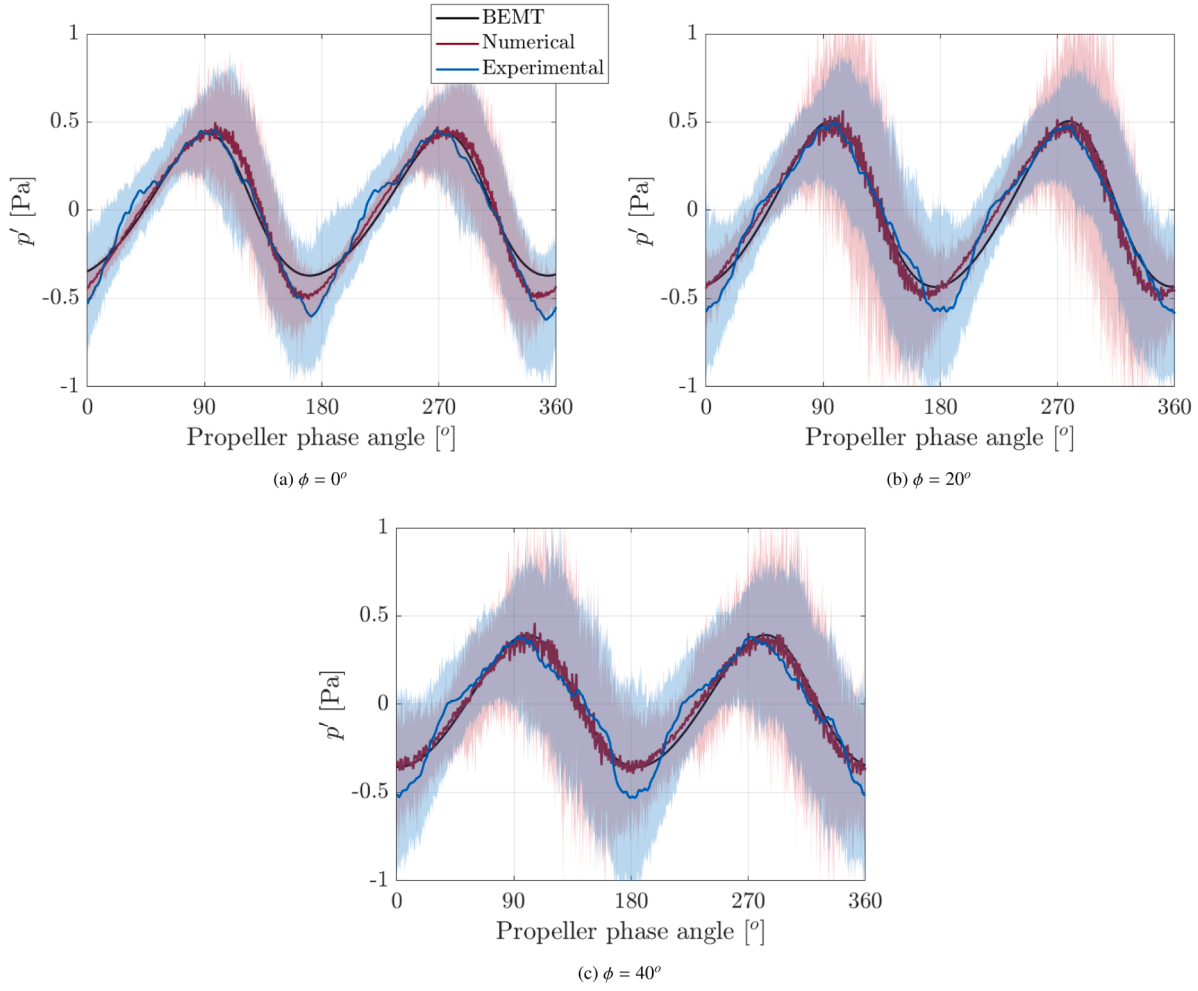


Fig. 7. Experimental, numerical, and BEMT-based average time history of the acoustic pressure fluctuations of the propeller at $\phi = 0^o$ obtained from phased-average procedure. Blue and red shades show maximum and minimum level deviation with respect to the pressure time history captured experimentally and numerically respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

verse of the number of rotations and is, for the numerical data, estimated in −17 dBs.

This is represented in the figure as a solid black line, for the analytical case, a red line, for the numerical one, and a blue line for the experimental measurements. The maximum deviations from the average time history, representing the maximum broadband or uncertainty components, are shown in shades of blue for the experiments and red for the numerical simulations. Following the previous observations of the noise spectrum, the agreement between the average pressure time histories follows the predictions of the first and second BPFs. Discrepancies are observed on the higher negative pressure peak from the experimental time history, around 0.2 Pa lower than the estimated one from the analytical method. Conversely, the analytical prediction results in a lower negative amplitude than the numerical and experimental results. Both are more evident for $\phi = 0^o$.

Between numerical simulations and experimental measurements, the level of the broadband fluctuations is well captured and the mean pressure signal is well in agreement [11]. Nevertheless, deviations exist. In particular, the broadband signal numerically predicted from the blades is higher when the average pressure has a negative derivative, i.e. broadband fluctuations increase when the tonal pressure signal is decreasing. This phenomenon indicates a modulation of the broadband signal, attributed to the dominance of trailing-edge noise, and its varying directivity as the propeller rotates [34]. The latter reference has observed, experimentally, the modulation effect over the same propeller at different conditions. In this study, the modulation is observed in the experimental data mildly. The reasons for that are the presence of stronger low-frequency broadband components, possibly related to spurious noise sources.

The directivity of the tonal sources is shown in Fig. 8 for the 6 first harmonics of the BPF. The data is obtained from the frequency analysis of the average time signal instead of the direct acoustic data. This methodology is selected as it creates a filter for the broadband signals, separating the tonal components only and allowing comparisons of the latter for higher harmonics. Predictions of the first two BPFs demonstrate agreement between the three methodologies, with larger discrepancies shown for the BEMT-based methodology. The latter demonstrates slightly lower levels towards the upstream direction, following the expected pattern from rotor noise theory [6]. This disagreement can be associated with slightly different combination of torque and thrust sources or the consideration of a single convective speed (inflow speed) along the whole path between the source and the microphone, opposite to the experimental, in which the airflow is restricted inside the jet stream of the wind tunnel nozzle.

Stronger differences are observed for the higher harmonics. Experimental results struggle with motor noise that affects the 3rd and 6th BPFs. Still, BPF harmonics above the 4th demonstrate large disagreement between all methodologies. BEMT-based predictions and numerical methods show similar levels while experiments consistently display significantly higher noise levels. In general, numerical methods and ana-

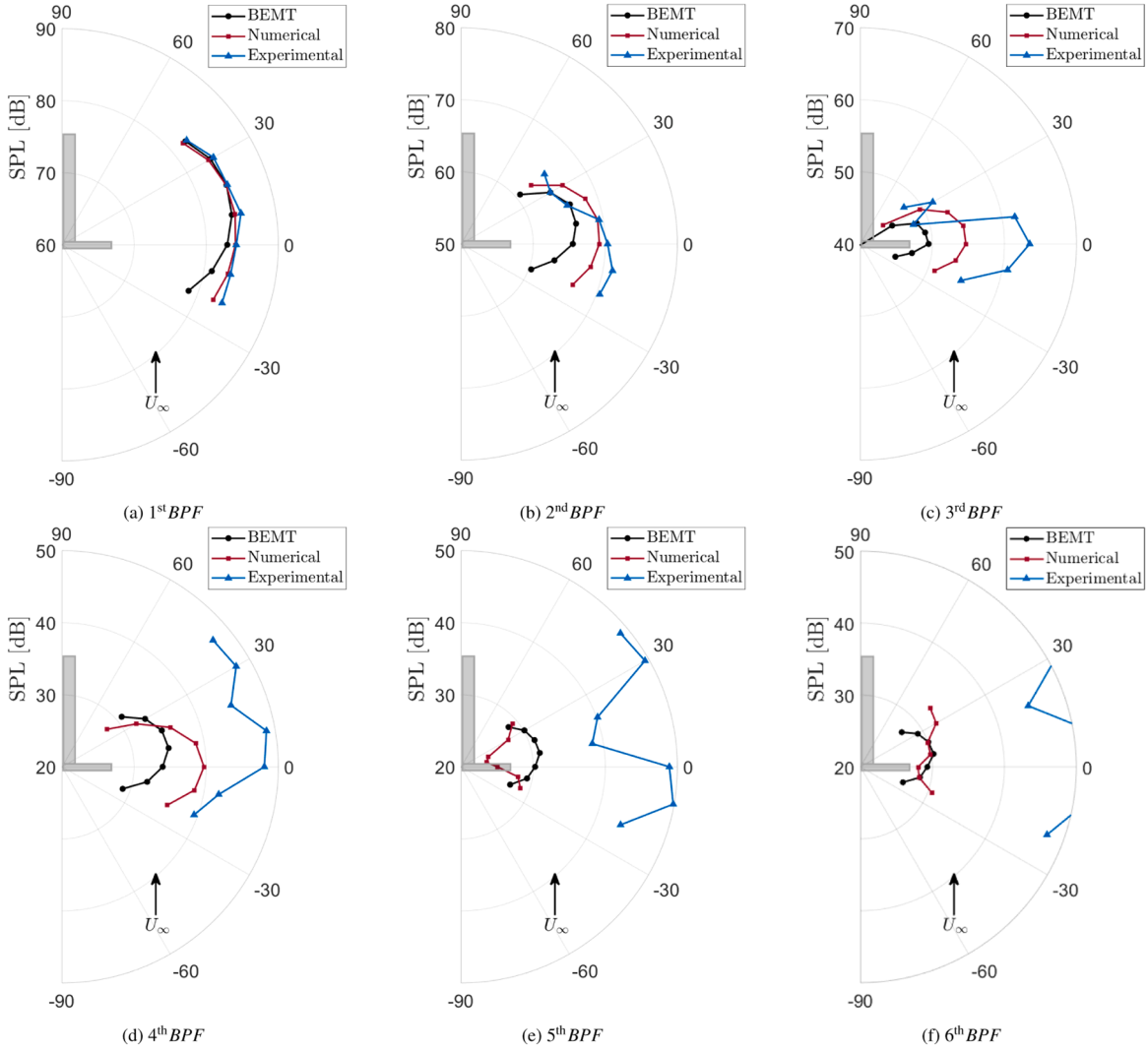


Fig. 8. Experimental, numerical, and BEMT-based phase-lock averaged time history of the acoustic pressure fluctuations of the propeller at $\phi = 0^o$.

lytical ones agree fairly well for the directivity patterns up to the 4th BPF harmonic. The three methodologies are able to predict the expected directivity patterns for an isolated propeller for small BPF harmonics (below the 4th harmonic) while significant differences are observed for the 5th and 6th ones. Notably, in this case, the analytical method consistently predicts higher levels along the propeller plane while the numerical one displays the opposite behaviour, i.e. higher levels emitted towards the propeller axis. This would indicate possible dominant contribution from thrust noise in the numerical predictions, opposite to the torque-based noise observed for the analytical ones.

Fig. 9 summarizes the analysis of the total tonal levels and the discrepancies with respect to the BEMT data. Here the BEMT is taken as a reference instead of the experimental uncertainties causing certain BPFs to exhibit significantly different levels. The picture shows clearly the progression of the discrepancies from the first two/three BPF harmonics to the others. In the first harmonic, errors are within ±1.7 dB range (below experimental uncertainties). The second and third harmonics see an increase in the errors demonstrated by the underprediction of the analytical methodology and the influence of the motor noise. Higher harmonics yield larger discrepancies, which are significantly lower for the numerical technique in comparison with the experimental one. Here, it must be mentioned that the tonal levels inferred from the experimental campaign are in range of 40 dB, similar to the ones directly observed in the acoustic spectrum in Fig. 6. The former can indicate that the levels obtained from the phase average are still contaminated by broadband noise or that higher tonal components exist in the experiment. The latter can possibly be caused by asymmetries, unsteady loading components, or other sources of motor noise.

Errors with respect to the numerical technique are dependent on the azimuthal location, and significantly different behaviours are observed between $\phi = 0^o$ and $\phi = 40^o$ for harmonic numbers above 6. These deviations have two possible sources. First, the load distribution predicted from the BEMT comes from simplifications, with required assumptions on the tridimensionality of the flow and the influence of viscous effects on the blade pressure distribution. These assumptions can lead to differences between the load distribution from the simulations and from the analytical methodology, hence contributing to different noise levels. A second possible source of error is the consideration of compact dipoles from the BEMT-based noise estimations. The latter should affect the high harmonics of the blade passing frequencies, where the blade airfoil geometry and load distribution along the chord can interfere with the noise emission pattern. It must be highlighted that the discrepancies illustrated in this figure correspond to a maximum frequency of around 2670 Hz (BPF-10), equivalent to an acoustic wavelength of around 0.127 m (around 12 times the blade tip chord). This points to the suitability of the compact dipole assumption on the estimation of the demonstrated tonal components. Further investigation, however, would

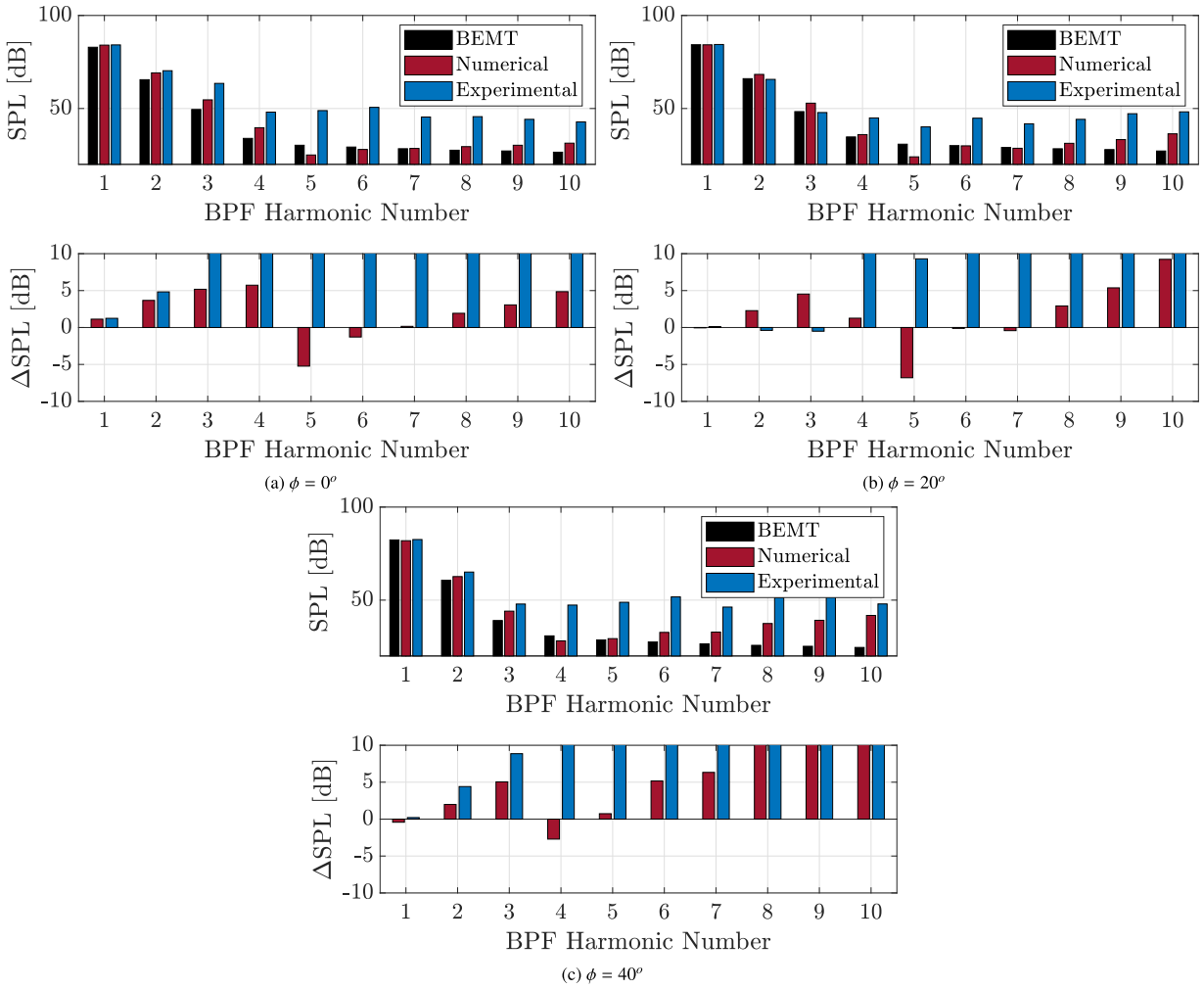


Fig. 9. Comparison of experimental, numerical, and BEMT-based phase-lock averaged tonal components at different harmonics of the blade passing frequency (BPF) of the propeller. Relative difference with respect to the BEMT-based estimation is also shown.

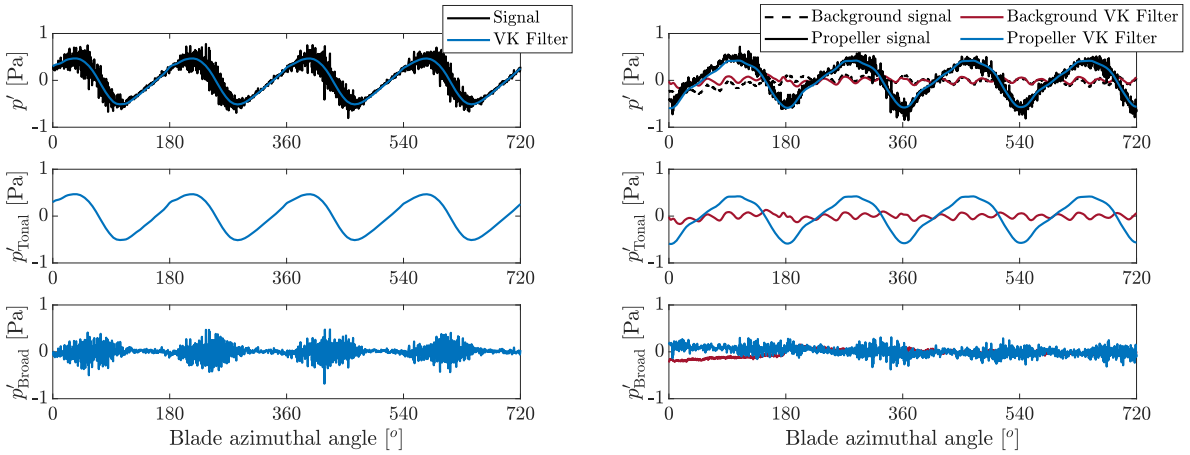
require a full assessment of the load distribution predicted from both methodologies and comparison with experimental results.

In summary, the analyses of the tonal components of the propeller noise point to the suitability of the three methodologies for the prediction of the first harmonics of the BPF. Those govern the overall noise emissions of this particular propeller and indicate that noise levels are well represented by the three possible methodologies. Nevertheless, an accurate description of the higher harmonics is lacking. Discrepancies are beyond 6 dB and vary with no recognizable pattern with different levels and directivities for the different methodologies. The following analysis will focus on the broadband emissions of the propeller.

6.3. Comparisons of broadband noise

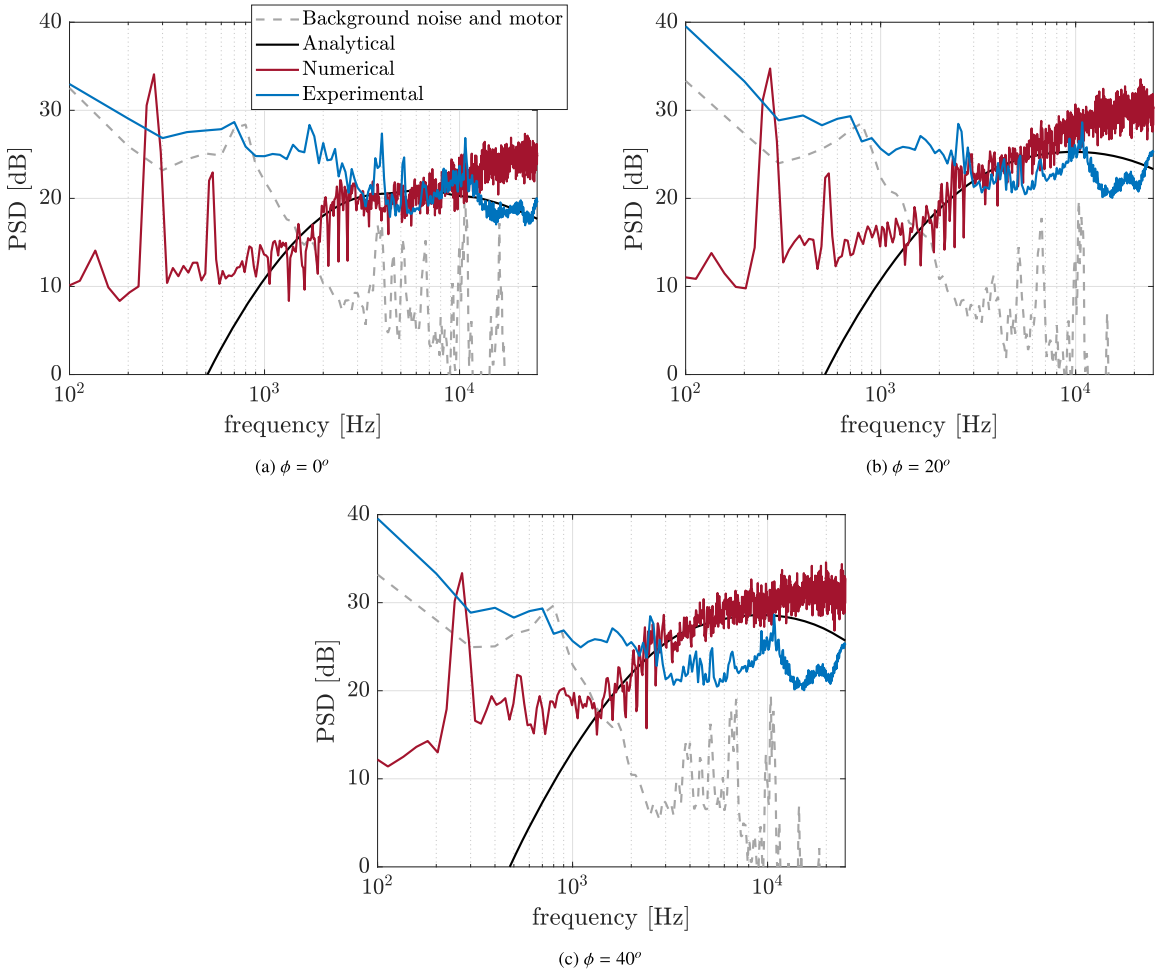
The level, spectrum, and directivity of the broadband noise component from the propeller are also compared as this component can dominate the emissions from a propeller operating at low-Reynolds numbers. To remove the tonal component from the broadband signal, a Vold-Kalman filter is applied [35]. A second-order filter is applied over a bandwidth of 20 Hz to avoid dampening the broadband modulation effects. Fig. 10 illustrates the effect of the filter on the three signals for an azimuthal angle of $\phi = 0^o$. The figure shows the broadband amplitude modulation described in Baars et al. [34]. The work also points to the relevance of the modulation effect on the noise perception. In this image, the modulation is observed both for the experimental and numerical data, with the former demonstrating lower peak levels and higher average levels of broadband fluctuations. This feature can have an important impact on the perception of noise and the significance of broadband noise propellers.

The spectra of the broadband fluctuations are shown in Fig. 11. Levels are now displayed in power spectral density (PSD), based on a reference density of 4×10^{-10} Pa^2/Hz, to allow direct comparison of broadband noise, independent of frequency band. At low frequencies, discrepancies are significantly large between BEMT-based, numerical, and experimental predictions. Tonal components are still remaining in the spectrum, especially in the numerical one, due to the lower number of rotational speeds modelled. Similarly, small tonal components, related to non-BPF harmonic frequencies, are still observed in the experimental broadband spectrum. Due to background and other possible sources in the experimental data, the levels for frequencies below 1 kHz display no agreement between acoustic emissions from analytical and numerical methods and experimental data. Above 2 kHz numerical, analytical, and experimental predictions are well in agreement. The agreement is best for the lowest azimuthal angle ($\phi = 0^o$). At higher azimuthal angles, experiments show lower levels than the analytical ones, while numerical models show higher levels. The numerical simulations show increasingly higher energies for frequencies above 10 kHz, indicating no decay of the turbulent fluctuations within the frequency range of interest, in contrast with analytical estimations and the experimental measurements. This could be due to some inaccuracies in the prediction of the laminar to turbulent transition or the presence of a laminar separation bubble, at-



(a) Numerical (b) Experimental

Fig. 10. Example of the filtered signal using the Vold-Kalman filtering technique. Time series are taken at azimuthal angle $\phi = 0^o$. (a) shows the results from filtering the numerical data and (b) the experimental one including the background noise of the motor-only measurements.



(a) $\phi = 0^o$ (b) $\phi = 20^o$

(c) $\phi = 40^o$

Fig. 11. Experimental, numerical, and BEMT-based Vold-Kalman filtered broadband power spectral density spectrum at different azimuthal angles (ϕ).

tributable to insufficient mesh resolution due to the blind nature of the present assessment, as pointed out for the same propeller operated at different advance ratios and rotational speed [5].

The directivity of the broadband noise emissions is evaluated in Fig. 12, where the levels are presented over octave bands derived from the Vold-Kalman filtered data [35]. A dipolar emission pattern, characterized by the lowest levels around the propeller plane is observed for most frequencies, indicating source compactness for the frequencies analysed. At frequencies lower than 2,000 Hz, experiments are dominated by background sources and are far above the predicted levels. Above this frequency, a better agreement between the three methodologies is seen. The agreement confirms the hypothesized dominance of

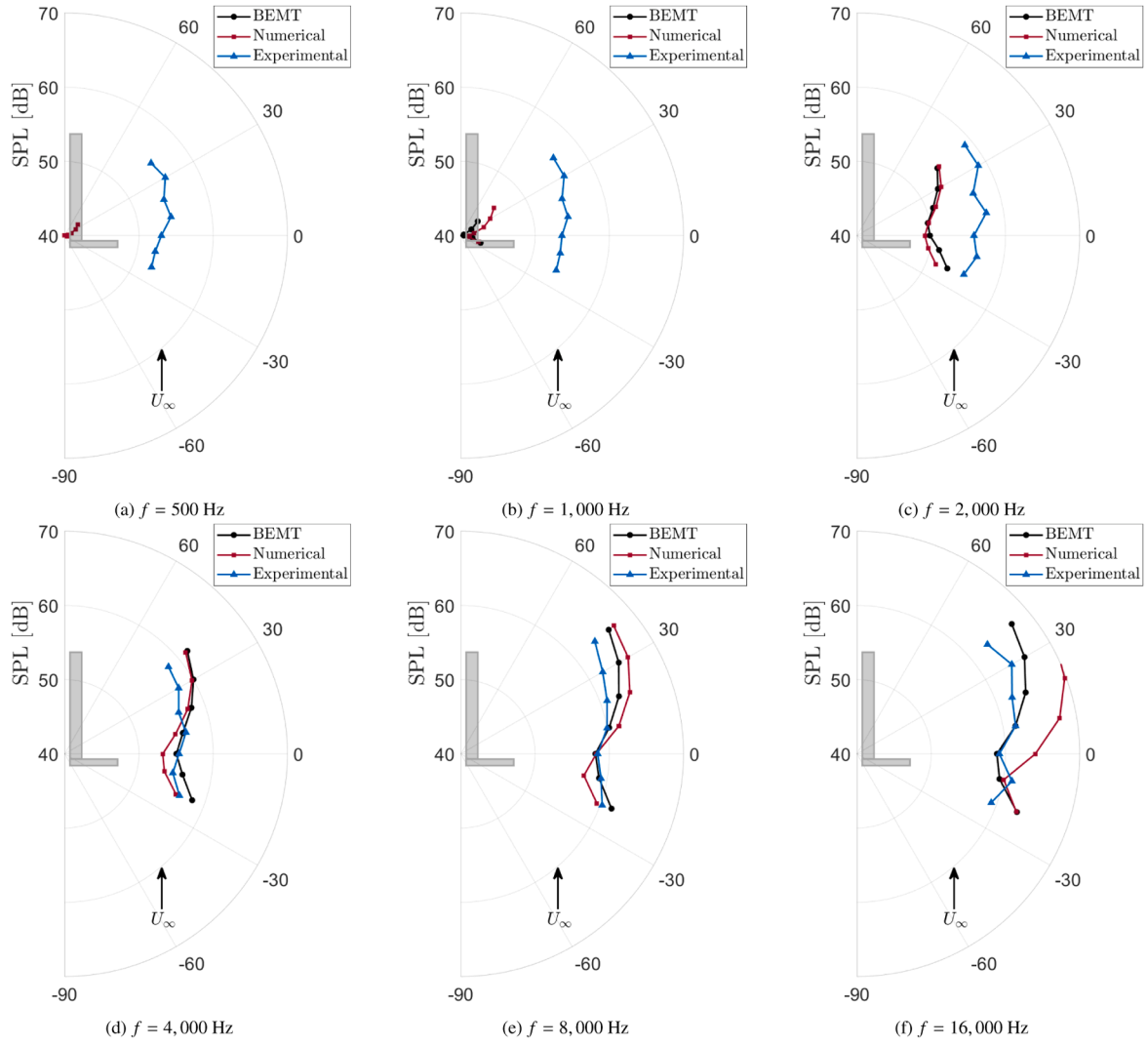


Fig. 12. Experimental, numerical, and BEMT-based directivity of the broadband noise emitted by the propeller at octave bands.

turbulent-boundary layer trailing-edge noise. Overall, discrepancies on broadband comparisons are well below 5 dB and are the highest for the azimuthal angle $\phi = 40^o$. At the highest frequency, the numerical simulation overestimates the noise emissions in comparison to both other methods. The discrepancies, however, are much slimmer when compared to the tonal components and point to a much more consistent prediction of broadband sources of noise at high frequencies.

7. Conclusions

This work focuses on a comparison of acoustic noise components from isolated propellers. A dedicated experimental campaign is carried out, and an equivalent numerical simulation and analytical prediction are conducted, following a blind test procedure. Focus is given to the currently achievable accuracy of high harmonics of the blade passing frequency and broadband sources for a specific condition of rotational speed and advance ratio. A discussion is proposed on the experimental uncertainties incurred from measurements of tonal propeller noise and the accuracy of numerical techniques, sources of discrepancies. Comparisons are shown, including a BEMT-based noise estimation, to propose a discussion on the level of accuracy of the tonal sound pressure levels and their time history along the blade rotation.

Results demonstrate that the three methods are accurate in the description of the first two BPF tonal components, within 3 dB precision and comparable to experimental limitations. These components commonly dominate the acoustic pressure fluctuations from propellers. As a result, the equivalent time history of pressure is well described by the three methodologies. Discrepancies, however, grow as the higher harmonics are to be considered. The latter demonstrates differences between analytical and experimental methods larger than 5 dB for all harmonics higher than the 3rd blade passing frequency of the blade. Numerical methods demonstrate better agreement with analytical predictions and discrepancies are only above 5 dB for harmonic numbers higher than 5. For the latter, errors are particularly higher for large azimuthal angles, where the directionality of the source is predicted inversely to the one suggested by the analytical methodology. Noise from the motors and test rigs is pointed out as one of the causes of the observed discrepancies from experimental measurements. Measurements of the rig without the propellers are shown, collaborating with the hypothesis for 3rd, 6th and 7th harmonic of the blade passing frequency. A final assessment of the sources of discrepancies is not possible due to the different torque requirements and lack of information on possible other sources, i.e. propeller and flow asymmetries.

Broadband signals are separated from the tonal ones with the use of Vold-Kalman filtering. Analytical, numerical, and experimental meth-

ods agree well in the highest frequency ranges (above 2,000 Hz). At such frequencies, all three methodologies display coherent levels and directivities. Discrepancies, in this case, are higher for the numerical methods, which indicate higher levels at high frequencies in comparison to the observed ones from the experiments and the ones predicted analytically. Deviations from the methodologies are larger at high frequencies and, mostly, follow the difficulties in predicting the turbulent flow conditions over the blade. Analytical methods depend strongly on the choice of wall-pressure spectrum model. Future work dedicated to maximizing the accuracy of broadband sources should look carefully at the choice of model.

In conclusion, this work illustrates the current comparison status of three established methodologies for assessing noise emissions from isolated propellers. The results shown demonstrate that overall noise levels are well-captured, in sub-dB-level precision, for the three methodologies. Nevertheless, a detailed spectral-based description of the emissions requires careful analysis of the results and management of uncertainties and sources of deviation. It is seen that discrepancies grow larger for the higher harmonics of the blade passing frequencies and can be well above 10 dB (one energy scale) between different methodologies. Besides, errors are shown in levels and directivity patterns, which point, in this case, to different dominant physics between each technique. Similar figures are observed for broadband emissions, where dipolar directivity patterns are displayed for all methodologies, indicating discrepancies in the turbulent levels over the blade. The discrepancies are of keen importance for applications of low-Reynolds numbers and low-RPM propellers, where increased relative levels of turbulence-based sources and lower human perception of low frequencies cause the higher harmonics of the blade passing frequency and the broadband sources of noise to dominate noise emissions. Following the latter, the importance of a more precise description of modelling conditions for numerical simulations, physical description of propeller sources, and, especially, carefully taken experimental techniques to suppress background and equipment noise sources and assess possible non-symmetric geometric and flow conditions at a deeper level of detail.

CRediT authorship contribution statement

Lourenço Tercio Lima Pereira: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Daniele Ragni:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Gianluca Romani:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Damiano Casalino:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

All the authors declare that they don't have any conflicts of interest.

Data availability

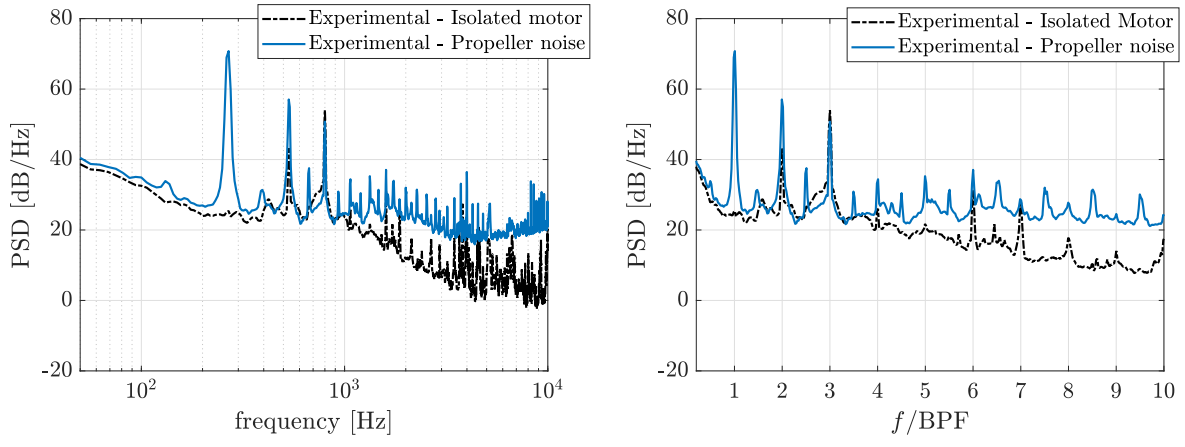
Data will be made available on request.

Appendix A. Experimental acoustic results and uncertainties

This appendix focuses on the experimental-only acoustic results. This is presented to create a more oriented discussion of some of the constraints of the experiments conducted and to understand the expected levels of uncertainties and limitations of the methodology. The spectrum of the propeller noise is shown in Fig. A.1 for the entire frequency range (a) and the details of the first 10 BPFs (b). The spectrum also shows the measured noise of the isolated motor operating at the same RPM. It is worth mentioning that the different torque demanded from the motor in the two conditions alters the noise emitted by the motor. Nevertheless, it can be observed that the motor noise is responsible for affecting significantly the 3rd, 6th and 7th BPF of the measured spectrum. The latter BPF harmonic is related to the number of poles from the motor used (14). The first two are, most probably, due to the 3 phase operation of brushless motors. Some sub-harmonic tones are also observed. Those can be caused by asymmetries on the propeller geometry or again by the noise of the isolated motor. By comparing the isolated motor component against the propeller measurement, one can observe that the sub-harmonics are mostly present in the propeller one. This points to the first hypothesis, i.e. asymmetries from the propeller or propeller mounting cause the sub-harmonic sources.

To better describe the tonal component of the noise from the propeller, Fig. A.2 shows the time history of pressure along one full rotation of the blade. The root-mean-square of the pressure fluctuations yields a tonal noise emission of 84.6 dB. Similarly, the pressure fluctuation of the isolated motor is also provided, which totals 67.4 dB. The latter is seen at a higher frequency, following the observations from its noise spectrum.

Experimental techniques are prone to uncertainties and accurate measurements of tonal noise are particularly challenging. To start, the microphones used deliver a flat frequency response, single sensitivity in all frequency ranges, with $\epsilon_{\text{mic.}} = \pm 1$ dB. This is the maximum accuracy achievable with the current setup.



(a) Logarithmic (b) Linear

Fig. A.1. Experimental acoustic spectrum of the propeller at $\phi = 0^o$ compared to the one of the isolated motor (a). (b) Details the first tonal components of the spectrum.

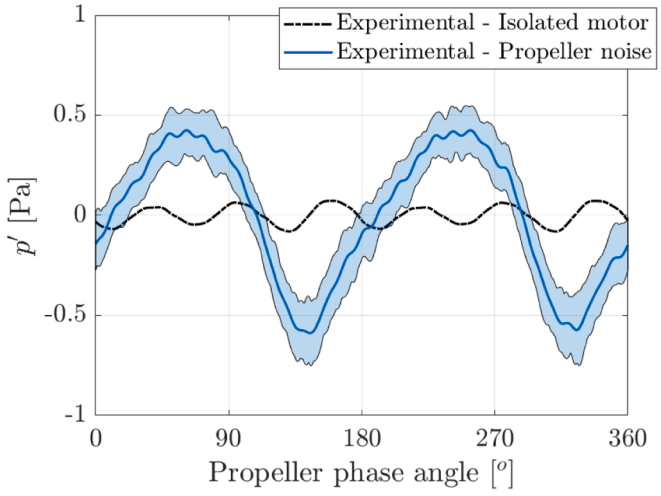


Fig. A.2. Experimental average time history of the acoustic pressure fluctuations of the propeller at $\phi = 0^o$ compared to the one of the isolated motor. Blue shades show maximum and minimum level deviation with respect to the pressure time-history acquired.

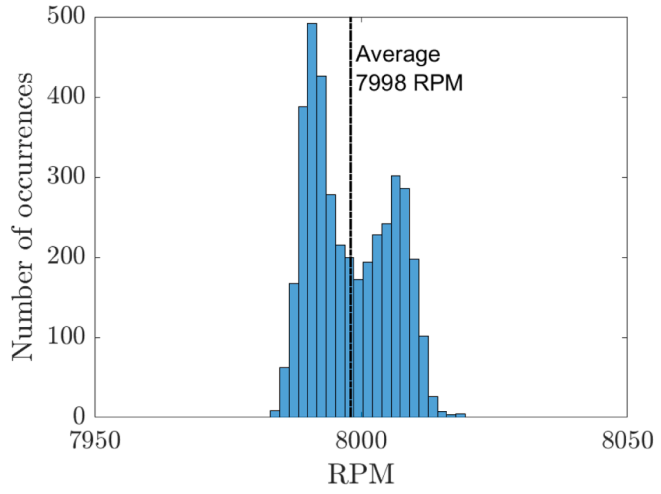


Fig. A.3. Experimental histogram of the RPM measured along the acoustic acquisition.

Besides the measuring equipment, controlling the RPM of the motor can be a cause of deviations. Fig. A.3 shows the histogram of RPMs captured during the data acquisition process. Overall, the RPM is controlled at an average of 7998 RPMs with a maximum dispersion of ±20 RPMs. By considering that tonal noise scales with the RPM according to a 6th power law, the error associated with the RPM variation is lower than $\epsilon_{\text{RPM}} = \pm 0.1$ dB.

Asymmetries in the flow or of the propeller geometry are another relevant source of uncertainties. The presence of non-uniform, unsteady force distribution over the blades significantly modifies the emissions from a propeller. The works of Goyal et al. and Monteiro et al. [36,37] have demonstrated how small modifications of the loading along a rotation can have a significant impact on the emitted noise. Within the average time-history of pressure fluctuations, the effect of asymmetry between the propellers can be inferred by the deviation of the time-history during each half of a rotation. Given the symmetry of the problem, the generated spectrum from the time history of the first half of rotation should be exactly the one of the second half. Deviations from that would indicate problems with the symmetry assumed on both blades. In the current experiment, the standard deviation of the time history between the first and second blade passing is 0.022 Pa, resulting in a possible change of the tonal noise levels of $\epsilon_{\text{geo.}} \pm 0.6$ dB (one standard deviation).

A final source of uncertainty to keep in mind is given by acoustic reflections inside the semi-anechoic chamber. The structures required for the setup, i.e. acoustic arc structure, propeller rig, nozzle, and accessories disturb the reflective properties of the anechoic chamber and, importantly, deviate from the simulated conditions. Reflections are responsible for possible constructive and destructive interference that can modify the measured levels. This source of error is hard to take into account as it can cause total cancellation or doubling of the source and, in this work, it is mitigated by increasing the spatial distribution of sensors and comparing results for all the azimuth angles measured.

In summary, it is extremely challenging to predict the uncertainty of microphone measurements, especially if reflections are to be considered. By neglecting the latter and applying error propagation theory, a deviation of ±1.7 dB is expected for the current experiment conducted. Nevertheless, this figure can be more limited when the spectral content is analysed. Due to motor noise interference, some of the harmonics of the motor shaft frequency are heavily influenced by the motor noise, hampering noise comparisons. It must be emphasized that motor noise is a critical issue to aeroacoustic propeller studies and future research should work on addressing this source of uncertainty. Besides, odd harmonics of the shaft frequency appear in the spectrum, which is not expected for a fully symmetric 2-bladed rotor and indicate propeller unbalance or manufacturing asymmetries. From this analysis, the comparison between the experimental and numerical techniques is carried out with a focus on discussing the discrepancies and possible sources of error.

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